

(6) 6.3

$$6. \hat{\mu} = \frac{1}{20} \sum_{i=1}^{20} \ln x_i = 3.089$$

$$\hat{\sigma}^2 = \frac{1}{20} \sum_{i=1}^{20} (\ln x_i - 3.089)^2 = 0.508$$

$$\therefore \hat{E}(X) = e^{\hat{\mu} + \frac{\hat{\sigma}^2}{2}} = 28.305$$

$$7. (1) EX = \frac{3\theta}{2} \quad DX = \frac{\theta^2}{12}$$

$$\therefore E\bar{X} = \frac{3\theta}{2} \quad D\bar{X} = \frac{\theta^2}{12n}$$

$$\therefore E\hat{\theta} = \theta$$

$$D\hat{\theta} = \frac{\theta^2}{27n} \rightarrow 0$$

$\therefore$ 无偏且相合

$$(2) L(\theta) = \left(\frac{1}{\theta}\right)^n I_{\{\theta < X_{(1)} < X_{(n)} < 2\theta\}}$$

当 $\hat{\theta} = \frac{X_{(n)}}{2}$ 时, $L(\theta)$ 有最大值 $\therefore$ 最大似然估计 $\hat{\theta} = \frac{X_{(n)}}{2}$

$$E\hat{\theta} = \frac{2n+1}{4(n+1)} \theta, \quad \therefore \text{不为无偏估计}$$

$$D\hat{\theta} = E\hat{\theta}^2 - (E\hat{\theta})^2 = \frac{n\theta^2}{2(1+2/(n+1))^2} \rightarrow 0 \quad \therefore \text{为相合估计}$$

$$8. (1) L(\theta) = e^{\theta \sum_{i=1}^n x_i - \frac{n}{2} \theta^2} I_{\{x_{(1)} > \theta\}}$$

当 $\theta = \bar{x}_{(1)}$ 时, 有最大值.

$$E \hat{\theta}_1 = \frac{1}{n} + \theta \neq \theta \quad \therefore \hat{\theta}_1 \text{ 为 } \theta \text{ 的有偏估计}$$

$$E \hat{\theta}_1^2 = \frac{2}{n^2} + \frac{2}{n} \theta + \theta^2$$

$$D \hat{\theta}_1 = \frac{1}{n^2} \rightarrow 0 \quad \therefore \hat{\theta}_1 \text{ 为相合估计}$$

$$(2) EX = \theta + 1 \quad \therefore \hat{\theta}_2 = \bar{X} - 1, \text{ 必为 } \theta \text{ 的无偏估计,}$$

$$D \hat{\theta}_2 = \frac{1}{n} \rightarrow 0$$

$\therefore \hat{\theta}_2$ 为 θ 的相合估计

6.4

2. 对任意 ϕ 满足 $E\phi = 0$, 有 $Cov(T_i, \phi) = 0$

$$E(aT_1 + bT_2) = a\theta_1 + b\theta_2$$

$$Cov(aT_1 + bT_2, \phi) = 0$$

$\therefore aT_1 + bT_2$ 为 $a\theta_1 + b\theta_2$ 的 UMVUE

$$6. (1) L(\theta) = \prod_{i=1}^n \theta x_i^{\theta-1} \cdot \theta^n$$

$$\frac{\partial \ln L(\theta)}{\partial \theta} = \cancel{\frac{n}{\theta}} - \frac{n}{\theta^2} - \frac{\sum_{i=1}^n \ln x_i}{\theta^2} = 0$$

$$\text{解得 } \hat{g}(\theta) = -\frac{1}{n} \sum_{i=1}^n \ln x_i$$

$$(2) I(\theta) = -E\left(\frac{\partial^2}{\partial \theta^2} \ln f(x, \theta)\right) = -E\left(-\frac{1}{\theta^2}\right) = \frac{1}{\theta^2}$$

$\therefore g(\theta)$ 的无偏估计 C-R 下界为 $\frac{1}{n\theta^2}$

$$\text{记 } Y_i = -\ln X_i$$

$$\text{则 } f_{Y_i}(y) = \theta e^{-\theta y} I_{\{y>0\}} \quad \therefore Y_i \sim \text{Exp}(\theta)$$

$$\therefore \hat{g}(\theta) \sim \text{Gamma}(n, n\theta)$$

$$\therefore E \hat{g}(\theta) = \frac{1}{\theta} \quad -D \hat{g}(\theta) = \frac{1}{n\theta^2}$$

$\therefore \hat{g}(\theta)$ 为 $g(\theta)$ 的有效估计

12. $T = \bar{X} \sim N(\mu, 1)$ 是一个充分完备统计量

且 $\bar{X}^2 - \frac{1}{n}$ 为 μ^2 的一个无偏估计.

$$E(\bar{X}^2 - \frac{1}{n} | \bar{X}) = \bar{X}^2 - \frac{1}{n}$$

由 Lehman-Scheffé 定理: $\bar{X}^2 - \frac{1}{n}$ 为 μ^2 的 UMVUE

$$\text{而 } D(\bar{X}^2 - \frac{1}{n}) = D(\bar{X}^2) = \frac{2^2}{n^2} + \frac{4}{n} \mu^2$$

$$C-R \text{ 下界} = \frac{(g'(\mu))^2}{nI(\mu)} = \frac{4\mu^2}{n}$$

即此 UMVUE 的方差达到 C-R 下界

$$(4) L(\theta) = \left(\frac{1-\theta}{2}\right)^{\frac{1}{2} \sum_{i=1}^n (x_i^2 - x_i)} \left(\frac{1}{2}\right)^{n - \sum_{i=1}^n x_i^2} \left(\frac{\theta}{2}\right)^{\frac{\sum_{i=1}^n x_i^2 + n}{2}}$$

$$\therefore \frac{\partial \ln L(\theta)}{\partial \theta} = -\frac{1}{2} \frac{\sum_{i=1}^n (x_i^2 - x_i)}{1-\theta} + \frac{1}{2\theta} \sum_{i=1}^n (x_i^2 + x_i) = 0$$

$$\text{解得: } \hat{\theta}_1 = \frac{1}{2} + \frac{\sum_{i=1}^n x_i}{2 \sum_{i=1}^n x_i^2}$$

$$E \hat{\theta}_1 = \frac{1}{2} + \frac{1}{2}(\theta - \frac{1}{2}) / (1 - \frac{1}{2n}) \neq 0 \quad \therefore \hat{\theta}_1 \text{ 为有偏估计}$$

$$(2) E X = \theta - \frac{1}{2} \quad \therefore \hat{\theta}_2 = \bar{X} + \frac{1}{2}$$

$$(3) I(\theta) = -E\left(\frac{\partial \ln f(x, \theta)}{\partial \theta}\right)^2 = \frac{1}{4\theta(1-\theta)}$$

$$\therefore \theta \text{ 的无偏估计的 C-R 下界为 } \frac{2\theta(1-\theta)}{n}$$

6.5

2. ~~$h(x, \theta) = \pi(\theta|x) m(x)$~~

$$h(x, \theta) = f(x|\theta) \pi(\theta) = \frac{1}{6}, 11.1 < \theta < 11.7$$

$$\pi(\theta|x) = \frac{h(x, \theta)}{\int_{-\infty}^{+\infty} h(x, \theta) d\theta} = \frac{5}{3}$$

∴ θ 的后验分布为 $U(11.1, 11.7)$

4. 若 λ 的先验分布为 ~~$\pi(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}$~~ $\lambda \sim \text{Gamma}(\alpha, \beta)$, $\pi(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}$

$$\begin{aligned} \pi(\lambda|x) &= \frac{f(x_1, x_2, \dots, x_n|\lambda) \pi(\lambda)}{\int_0^{+\infty} f(x_1, x_2, \dots, x_n|\lambda) \pi(\lambda) d\lambda} \\ &= \frac{(\beta+n)^{\sum_{i=1}^n x_i + \alpha}}{\Gamma(\sum_{i=1}^n x_i + \alpha)} \lambda^{\sum_{i=1}^n x_i + \alpha - 1} e^{-(\beta+n)\lambda} \sim \text{Gamma}(\sum_{i=1}^n x_i + \alpha, \beta+n) \end{aligned}$$

∴ 伽马分布为共轭先验分布

$$7. \pi(\theta|x) = \frac{f(x_1, x_2, \dots, x_n|\theta) \pi(\theta)}{\int_0^{+\infty} f(x_1, x_2, \dots, x_n|\theta) \pi(\theta) d\theta} = \frac{(\lambda - \sum_{i=1}^n \ln x_i)^{n+\alpha}}{\Gamma(\lambda - \sum_{i=1}^n \ln x_i)} \theta^{n+\alpha-1} e^{-\theta(\lambda - \sum_{i=1}^n \ln x_i)}$$

∴ θ 的后验分布为 $\Gamma(n+\alpha, \lambda - \sum_{i=1}^n \ln x_i)$

$$\hat{\theta}_B = \frac{n+\alpha}{\lambda - \sum_{i=1}^n \ln x_i}$$

$$8. (1) \pi(\theta|x) = \frac{f(x_1, x_2, \dots, x_n|\theta) \pi(\theta)}{\int_{\max\{x_n, \theta_0\}}^{+\infty} f(x_1, x_2, \dots, x_n|\theta) \pi(\theta) d\theta} = \frac{(n+\beta) \theta (\max\{x_n, \theta_0\})^{n+\beta}}{\theta^{n+\beta+1}}$$

此为参数为 $n+\beta+1$ 和 $\max\{x_n, \theta_0\}$ 的帕累托分布.

$\therefore$ 帕累托分布为共轭先验分布

$$(2) \hat{\theta}_\beta = \hat{\theta} = \int_{\max\{x_n, \theta_0\}}^{+\infty} \theta \cdot \pi(\theta|x) d\theta = \frac{(n+\beta) \max\{x_n, \theta_0\}}{n+\beta-1}$$